

torch.index_select#

https://pytorch.org/docs/stable/generated/torch.index_select.html#torch.index_select

Description:

Returns a new tensor which indexes the input tensor along dimension `dim` using the entries in `index` which is a `LongTensor`. The returned tensor has the same number of dimensions as the original tensor (input). The `dimth` dimension has the same size as the length of `index`; other dimensions have the same size as in the original tensor. The returned tensor does not use the same storage as the original tensor. If `out` has a different shape than expected, we silently change it to the correct shape, reallocating the underlying storage if necessary. `input (Tensor)` – the input tensor.

Function Signature

```
torch.index_select(input, dim, index, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> x = torch.randn(3, 4)
>>> x
tensor([[ 0.1427,  0.0231, -0.5414, -1.0009],
        [-0.4664,  0.2647, -0.1228, -1.1068],
        [-1.1734, -0.6571,  0.7230, -0.6004]])
>>> indices = torch.tensor([0, 2])
>>> torch.index_select(x, 0, indices)
tensor([[ 0.1427,  0.0231, -0.5414, -1.0009],
        [-1.1734, -0.6571,  0.7230, -0.6004]])
>>> torch.index_select(x, 1, indices)
tensor([[ 0.1427, -0.5414],
        [-0.4664, -0.1228],
        [-1.1734,  0.7230]])
```

Notes & Warnings

NOTE

Note The returned tensor does not use the same storage as the original tensor. If out has a different shape than expected, we silently change it to the correct shape, reallocating the underlying storage if necessary.

Detailed Documentation

Documentation

Returns a new tensor which indexes the input tensor along dimension `dim` using the entries in `index` which is a `LongTensor`.

The returned tensor has the same number of dimensions as the original tensor (input). The `dim`th dimension has the same size as the length of `index`; other dimensions have the same size as in the original tensor.

The returned tensor does not use the same storage as the original tensor. If `out` has a different shape than expected, we silently change it to the correct shape, reallocating the underlying storage if necessary.

`input (Tensor)` – the input tensor.

`dim (int)` – the dimension in which we index

`index (IntTensor or LongTensor)` – the 1-D tensor containing the indices to index

`out (Tensor, optional)` – the output tensor.